

OECD 497 Defined Approach - Regulatory Compliance Dossier

Generated by Skin Sensitizer AI (SSai) • 3D Quantum Mechanics & NAM Engine

1. Substance Identification & Hazard Call

Parameter	Value / Evaluation
Substance Name	Cinnamaldehyde
SMILES Notation	O=CC=Cc1ccccc1
AOP Key Event 1 Call	SENSITIZER (Category 1)
Validation Status	Verified (Out-of-sample Tier-1 & Massive Suite)

2. In Vitro NAM Readouts & Mechanistic Domain

Assay / Module	Predicted Outcome / Endpoint
Direct Peptide Reactivity (DPRA)	Positive (Cysteine & Lysine peptide depletion > 25%)
KeratinoSens (ARE-Nrf2)	Positive (Luciferase induction > 1.5-fold, EC1.5 < 1000 uM)
Structural Alert Domain	Alpha,beta-unsaturated aldehyde / Michael acceptor

3. 3D Quantum-Chemical & Thermodynamic Profile (\$E_{LUMO}\$)

Quantum Descriptor	Calculated Value / Interpretation
Calculated LUMO Energy	-1.42 eV (Indicates strong electron-accepting reactivity)
Electrophilicity Index (\$\omega\$)	0.73 (Quantifies global electrophilic power)
Thermodynamic Verdict	REACTIVE ELECTROPHILE (Confirmed favorable for protein-binding)

4. Regulatory Quality Assurance

This dossier was processed through SSai's multi-tier validation architecture, combining 3D conformer generation (ETKDG), semi-empirical quantum orbital calculations, and robust QSAR rules aligned with OECD 497 guidelines.